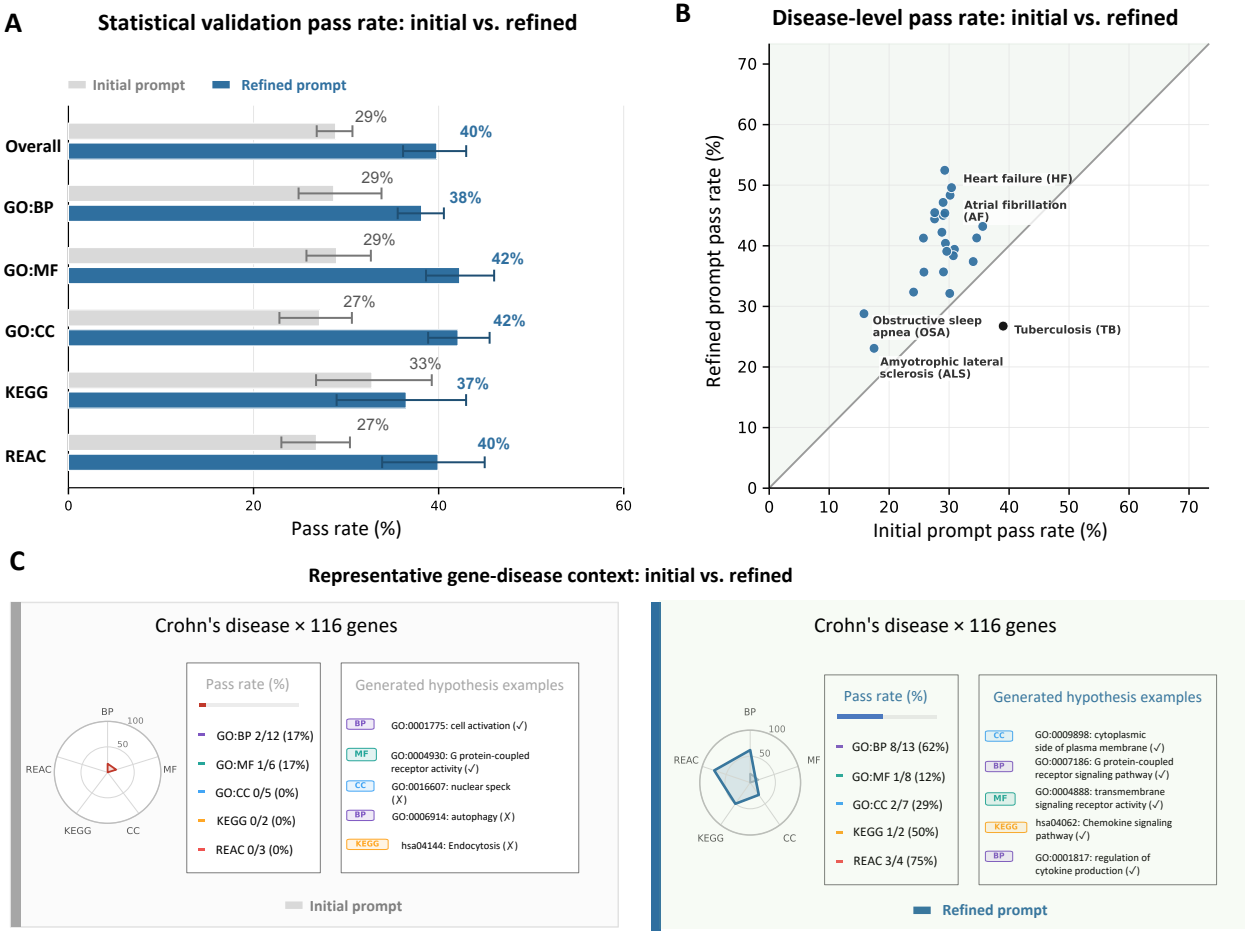

A
Statistical validation pass rate: initial vs. refined
Initial prompt
Refined prompt
Overall
GO:BP
GO:MF
GO:CC
KEGG
REAC
29%
40%
29%
38%
29%
42%
27%
42%
33%
37%
27%
40%
0
20
40
60
Pass rate (%)
B
Disease-level pass rate: initial vs. refined
Refined prompt pass rate (%)
Initial prompt pass rate (%)
Heart failure (HF)
Atrial fibrillation (AF)
Obstructive sleep apnea (OSA)
Tuberculosis (TB)
Amyotrophic lateral sclerosis (ALS)
C
Representative gene-disease context: initial vs. refined
Crohn's disease × 116 genes
Pass rate (%)
GO:BP 2/12 (17%)
GO:MF 1/6 (17%)
GO:CC 0/5 (0%)
KEGG 0/2 (0%)
REAC 0/3 (0%)
Generated hypothesis examples
GO:0001775: cell activation (✓)
GO:0004930: G protein-coupled receptor activity (✓)
GO:0016607: nuclear speck (X)
GO:0006914: autophagy (X)
hsa04144: Endocytosis (X)
Initial prompt
Crohn's disease × 116 genes
Pass rate (%)
GO:BP 8/13 (62%)
GO:MF 1/8 (12%)
GO:CC 2/7 (29%)
KEGG 1/2 (50%)
REAC 3/4 (75%)
Generated hypothesis examples
GO:0009898: cytoplasmic side of plasma membrane (✓)
GO:0007186: G protein-coupled receptor signaling pathway (✓)
GO:0004888: transmembrane signaling receptor activity (✓)
hsa04062: Chemokine signaling pathway (✓)
GO:0001817: regulation of cytokine production (✓)
Refined prompt

Figure 3